

Subject to:

$$\begin{array}{ll}
 r_{i1} = X_i & \text{for all } i \\
 r_{in+1} = 0 & \text{for all } i \\
 0 \leq r_{ij} - r_{ij+1} \leq x_{i,\max}^* & \text{for all } i, j \\
 r_{ij} \geq 0 & \text{for all } i, j
 \end{array}$$

Notice that this formulation is written only in terms of the residual shares. This is permissible since $x_{ij} = r_{ij} - r_{ij+1}$. Additionally, $r_{i1} = X_i$ is the proper residual starting value and $r_{in+1} = 0$ is the terminal value to ensure all shares are transacted by the end of trading and satisfying our completion requirement. The last two constraints ensure the solution adheres to the shrinking portfolio constraint, and the minimum and maximum trade quantity values. Finally, λ^* is the variance aversion parameter and is different from the risk aversion parameter.

An inherent difficulty with the QP solution, however, is that there is no way to map risk aversion to variance aversion, so the actual process may need several iterations to determine the solution at the desired level of risk aversion.

Another difficulty is that the formulated problem dramatically increases in size as the number of stocks in the portfolio increases. This may diminish the efficiency benefits of the QP approach as the trade list becomes too large.

In matrix notation, the quadratic optimization is written as follows:

$$\text{Min}_{\tilde{r}} \frac{1}{2} \cdot \tilde{r}' Q \tilde{r} \quad (9.7)$$

Subject to:

$$\begin{array}{l}
 \tilde{A}_1 \tilde{r} = \tilde{b}_1 \\
 \tilde{A}_2 \tilde{r} \geq \tilde{b}_2 \\
 \tilde{r}_{ij} \geq 0
 \end{array}$$

Where, $\tilde{r} = m \cdot (n+1) \times 1$, $Q = m \cdot (n+1) \times m \cdot (n+1)$, $\tilde{A}_1 = 2m \times m \cdot (n+1)$, $\tilde{b}_1 = 2m \times 1$, $\tilde{A}_2 = 2m \cdot (n+1) \times m \cdot (n+1)$, and $\tilde{b}_2 = 2m \cdot (n+1) \times 1$. The derivation of these matrices is provided in the appendix to this chapter.

This representation of the trader's dilemma above provides many advantages. First, there are many well known optimization algorithms suited to solve a QP minimization problem. Second, this formulation allows us to take complete advantage of diversification and hedging opportunities.